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ROLL NO: 49

SUBJECT: AOA

EXPERIMENT NO: 6

To implement Prim's MST Algorithm using Greedy Method.

Code

```
#include <stdio.h>

#include <limits.h>


#define MAX_VERTICES 100


int minKey(int key[], int mstSet[], int vertices) {

    int min = INT_MAX;

    int min_index = -1;


    for (int v = 0; v < vertices; ++v) {

        if (!mstSet[v] && key[v] < min) {

            min = key[v];

            min_index = v;

        }

    }

    return min_index;
```

```
}

void primMST(int graph[MAX_VERTICES][MAX_VERTICES], int vertices) {
    int parent[MAX_VERTICES];
    int key[MAX_VERTICES];
    int mstSet[MAX_VERTICES];

    for (int i = 0; i < vertices; ++i) {
        key[i] = INT_MAX;
        mstSet[i] = 0;
    }

    key[0] = 0;
    parent[0] = -1;

    for (int count = 0; count < vertices - 1; ++count) {
        int u = minKey(key, mstSet, vertices);
        mstSet[u] = 1;

        for (int v = 0; v < vertices; ++v) {
            if (graph[u][v] && !mstSet[v] && graph[u][v] < key[v]) {
                parent[v] = u;
```

```
            key[v] = graph[u][v];
        }
    }
}

    // Print the MST
    printf("Edge \tWeight\n");
    for (int i = 1; i < vertices; ++i) {
        printf("%d - %d\t%d\n", parent[i], i, graph[i][parent[i]]);
    }
}

int main() {
    int vertices;
    printf("Enter the number of vertices: ");
    scanf("%d", &vertices);

    int graph[MAX_VERTICES][MAX_VERTICES];
    printf("Input the adjacency matrix for the graph:\n");
    for (int i = 0; i < vertices; ++i) {
        for (int j = 0; j < vertices; ++j) {
            scanf("%d", &graph[i][j]);
```

```
        }
    }

    primMST(graph, vertices);
    return 0;
}
```

Output:

```
Enter the number of vertices: 5
Input the adjacency matrix for the graph:
0
2
0
6
0
2
0
3
8
5
0
3
0
0
7
6
8
0
0
9
0
5
7
9
0
Edge     Weight
0 - 1    2
1 - 2    3
0 - 3    6
1 - 4    5
```